

Article

Real-Time Sensorless Speed Control of PMSMs Using a Runge–Kutta Extended Kalman Filter

Adile Akpunar Bozkurt 

Mechatronics Engineering Department, Faculty of Technology, Pamukkale University, 20160 Denizli, Türkiye; aakpunar@pau.edu.tr; Tel.: +90-532-491-98-76

Abstract

Permanent magnet synchronous motors (PMSMs) are widely preferred in modern applications due to their high efficiency, high torque-to-inertia ratio, high power factor, and rapid dynamic response. Achieving optimal PMSM performance requires precise control, which depends on accurate estimation of motor speed and rotor position. This information is traditionally obtained through sensors such as encoders; however, these devices increase system cost and introduce size and integration constraints, limiting their use in many PMSM-based applications. To overcome these limitations, sensorless control strategies have gained significant attention. Since PMSMs inherently exhibit nonlinear dynamic behavior, accurate modeling of these nonlinearities is essential for reliable sensorless operation. In this study, a Runge–Kutta Extended Kalman Filter (RKEKF) approach is developed and implemented to enhance estimation accuracy for both rotor position and speed. The developed method utilizes the applied stator voltages and measured phase currents to estimate the motor states. Experimental validation was conducted on the dSPACE DS1104 platform under various operating conditions, including forward and reverse rotation, acceleration, low- and high-speed operation, and loaded operation. Furthermore, the performance of the developed RKEKF under load was compared with the conventional Extended Kalman Filter (EKF), demonstrating its improved estimation capability. The real-time feasibility of the developed RKEKF was experimentally verified through execution-time measurements on the dSPACE DS1104 platform, where the conventional EKF and the RKEKF required 47 μ s and 55 μ s, respectively, confirming that the proposed approach remains suitable for real-time PMSM control while accommodating the additional computational effort associated with Runge–Kutta integration.

Keywords: permanent magnet synchronous motor (PMSM); sensorless control; Extended Kalman Filter (EKF); real-time implementation; state estimation

MSC: 93C10; 93E11; 93C57



Academic Editor: Zhijia Zhao

Received: 7 December 2025

Revised: 28 December 2025

Accepted: 30 December 2025

Published: 12 January 2026

Copyright: © 2026 by the author.

Licensee MDPI, Basel, Switzerland.

This article is an open access article distributed under the terms and conditions of the [Creative Commons Attribution \(CC BY\)](https://creativecommons.org/licenses/by/4.0/) license.

1. Introduction

PMSMs have been increasingly employed in various applications, particularly in industrial settings, due to their superior features, including high efficiency, low inertia, and high power density. However, controlling PMSMs presents significant challenges because of their nonlinear dynamic behavior and sensitivity to parameter variations [1]. In terms of control strategies, although linear control methods such as proportional–integral–derivative (PID) controllers are widely used in industrial applications, their performance often falls

short when addressing the nonlinear characteristics of PMSMs. Consequently, numerous nonlinear control techniques, such as model predictive control [2,3], sliding mode control [4,5], fuzzy logic control [6], and adaptive speed control [7], have been extensively investigated. Effective PMSM control requires accurate measurement of variables such as speed and rotor position to facilitate feedback control [8]. Traditionally, speed sensors or optical encoders are integrated into PMSMs for this purpose [9]. However, these sensors have several disadvantages, including increased cost, added volume and inertia, and reduced system reliability [10]. These limitations have shifted the focus toward sensorless control methods, which estimate rotor speed and position by leveraging measurements of stator terminal quantities [11]. In addition, enhanced sliding-mode observer-based vector control algorithms have been proposed for medium- and high-speed PMSM applications [12]. For example, fuzzy logic-based sliding mode observers have proven effective in rotor position estimation for sensorless control [13]. Similarly, neural network observer-based sensorless speed control methods have achieved promising results [14]. Another widely studied approach is the Model Reference Adaptive System (MRAS), which estimates rotor position and speed by minimizing the error between actual and estimated rotor positions [15].

A review of the literature on sensorless control methods highlights the increasing prominence of observer-based techniques. While linear observers, such as Luenberger observers and Kalman filters, are commonly employed, their estimation performance often falls short in addressing the nonlinear dynamics of PMSMs. As a result, advanced nonlinear estimation approaches have been favored. Among these, the Extended Kalman Filter (EKF) has emerged as a significant solution [16]. In the presence of measurement and system noise, as well as parameter variations, accurately estimating the speed is a highly challenging task. In cases where parameter variations and measurement noise are present, the Kalman Filter (KF) provides an ideal solution for estimating the speed, torque, and stator resistance of the PMSM [17]. The Kalman filter is extensively used as a stochastic observer, capable of providing real-time state estimation by combining measured states with noisy signal covariances [18]. In general, the EKF method is a more attractive and popular approach due to its ability to provide fast, precise, and accurate estimations. Thanks to these features, it is widely used in both research studies and practical applications. EKF is an iterative algorithm based on minimum variance and is designed using the mathematical model of the system. However, this approach introduces specific computational challenges. In recent years, the development of high-speed processors, such as Field-Programmable Gate Arrays (FPGAs) and Digital Signal Processors (DSPs), has significantly reduced the computational burden associated with EKF, facilitating its widespread use in sensorless vector control systems. EKF has a wide speed range and high immunity compared to other observer algorithms, making it a crucial method in research on motor state estimation [19]. EKF has been implemented for the sensorless operation of PMSMs using an ARM Cortex-M3 microcontroller (ARM Ltd., Cambridge, UK) [20]. Sensorless control of PMSMs has also been achieved using the EKF algorithm together with a four-switch three-phase inverter [21]. Field-Oriented Control (FOC) has been applied to PMSMs, enabling speed estimation using the Extended Kalman Filter (EKF) without relying on mechanical parameters. In one of these studies, the control process was implemented in real time using the TMS320F28379D board (Texas Instruments Inc., Dallas, TX, USA) [22]. In research on sensorless PMSM drives, methods combining EKF for medium and high-speed regions with low-frequency signal injection (LFSI) estimators for the low-speed region have been proposed. This approach has been implemented on a Field-Programmable Gate Array (FPGA) board [23]. Additionally, studies have demonstrated that the speed and rotor position of PMSM can be estimated using an Adaptive EKF, and the PMSM can be operated in a sensorless manner via an FPGA chip [24]. Furthermore, in PMSM systems powered

by a photovoltaic (PV) array and used to drive water pumps, speed and rotor position estimation have been performed using EKF, while the control process has been carried out with the TMS320C31 DSP (Texas Instruments Inc., Dallas, TX, USA) [25]. Recently, Runge–Kutta-based improvements to the EKF framework have also been investigated for PMSM sensorless control. In [26], the impact of Euler-based discretization on EKF estimation accuracy is analyzed, and a Runge–Kutta-based EKF is proposed to reduce position estimation errors, with additional enhancements to improve performance during acceleration. The validation of the proposed approach is limited to MATLAB/Simulink R2025b (MathWorks Inc., Natick, MA, USA)-based simulation studies.

In this study, an RKEKF, initially described in [27] and designed for nonlinear systems, is proposed to estimate the rotor speed and position of PMSMs for sensorless operation. Specifically, the fourth-order RKEKF numerical integration method was adapted and optimized for the real-time sensorless control of PMSMs. To evaluate its applicability, the proposed method was implemented on the dSPACE DS1104 hardware platform (dSPACE GmbH, Paderborn, Germany). The algorithm's computational efficiency was confirmed through an experimentally measured execution time of 55 μ s, demonstrating its suitability for real-time motor control applications. The performance of the RKEKF method was thoroughly evaluated under various operating conditions, including forward and reverse rotation, acceleration, low-speed and high-speed operation, and loaded operation. The results verified the method's accuracy in position estimation and its robustness in maintaining stable motor operation across different scenarios. Furthermore, a comparative analysis was conducted with the conventional EKF method under loaded conditions, clearly highlighting the advantages of the RKEKF approach. Overall, the findings confirm that the proposed method is suitable for use in fast dynamic systems such as electric drives. The contribution of this study lies in the experimental investigation of the real-time applicability of a Runge–Kutta-based Extended Kalman Filter for sensorless PMSM control. This work emphasizes practical implementation aspects, including execution-time analysis and computational cost evaluation. The proposed RKEKF is implemented on a dSPACE DS1104 platform and directly compared with the conventional EKF under various operating conditions, thereby providing quantitative insights into the trade-off between numerical integration accuracy and real-time feasibility.

The remainder of the study is structured as follows: Section 2 details PMSM models; Section 3 presents EKF fundamentals; Section 4 describes the RKEKF algorithm; Section 5 provides experimental results and comparisons.

2. PMSM Mathematical Equations

When the stator currents (i_α, i_β), the mechanical angular velocity (ω_r), and the electrical rotor position (θ_e) are defined as the state variables, the dynamic model of the Permanent Magnet Synchronous Motor (PMSM) in the stationary orthogonal reference frame (α – β) can be formulated as given in Equation (1):

$$\begin{aligned} \frac{di_\alpha}{dt} &= -\frac{R_s}{L_s} i_\alpha + \frac{\lambda p \omega_r \sin(\theta_e)}{L_s} + \frac{U_\alpha}{L_s} \\ \frac{di_\beta}{dt} &= -\frac{R_s}{L_s} i_\beta + \frac{\lambda p \omega_r \cos(\theta_e)}{L_s} + \frac{U_\beta}{L_s} \\ \frac{d\omega_r}{dt} &= 0 \\ \frac{d\theta_e}{dt} &= p \omega_r \end{aligned} \quad (1)$$

The state vector, input vector, and output vector of the PMSM system are, respectively, expressed as: $x = [i_\alpha \ i_\beta \ \omega_r \ \theta_e]^T$, $u = [U_\alpha \ U_\beta]^T$, $y = [i_\alpha \ i_\beta]^T$.

In these equations, R_s denotes the stator resistance, L_s represents the stator inductance, λ_p is the permanent magnet flux linkage, p indicates the number of pole pairs, and U_α , U_β correspond to the stator input voltages [28].

As the motor speed changes continuously during acceleration, ignoring the speed variation may degrade estimation accuracy. To address this issue, the rotor dynamics equation is adopted to capture the relationship between the electrical angle and speed, as shown in Equation (2) [26]:

$$\begin{aligned}\frac{di_\alpha}{dt} &= -\frac{R_s}{L_s}i_\alpha + \frac{\lambda_p\omega_r\sin(\theta_e)}{L_s} + \frac{U_\alpha}{L_s} \\ \frac{di_\beta}{dt} &= -\frac{R_s}{L_s}i_\beta + \frac{\lambda_p\omega_r\cos(\theta_e)}{L_s} + \frac{U_\beta}{L_s} \\ \frac{d\omega_r}{dt} &= \frac{3}{2}\frac{\lambda}{J}(-i_\alpha\sin\theta_e + i_\beta\cos\theta_e) - \frac{T_L}{J} \\ \frac{d\theta_e}{dt} &= p\omega_r\end{aligned}\quad (2)$$

3. Extended Kalman Filter (EKF)

The Kalman filter is a recursive mathematical algorithm that estimates the internal states of a linear dynamic system by operating in parallel with the real process. It continuously updates its predictions by comparing the measured outputs with the estimated ones, thereby minimizing the estimation error through an optimal feedback gain. The block diagram of a standard Extended Kalman Filter is illustrated in Figure 1.

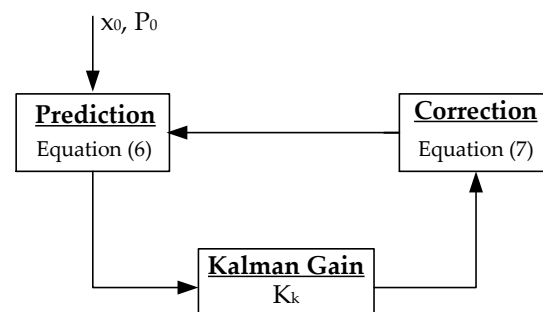


Figure 1. Block diagram of the EKF.

In real-world control applications, most processes exhibit nonlinear behavior. To address such cases, the Extended Kalman Filter (EKF) was developed as an enhanced version of the classical Kalman filter. EKF applies the principles of the least squares estimation method to nonlinear dynamic systems, providing an optimal solution under Gaussian noise assumptions.

The discrete-time mathematical representation of a nonlinear system can be expressed as shown in Equation (3) [20,29].

$$\begin{aligned}x_k &= f(x_{k-1}, u_k) + w_k \\ y_k &= h(x_k) + v_k\end{aligned}\quad (3)$$

here, k denotes the current sampling instant, while $k + 1$ corresponds to the subsequent time step, w_k and v_k denote the process and measurement noise, respectively. Both noise components are assumed to be zero-mean, white, Gaussian, and mutually uncorrelated. In the conventional EKF framework, the state equation is discretized and linearized using the Euler approximation, as given in Equation (4).

$$\begin{aligned}F_k &= 1 + f(x_k, u)T \\ y_k &= h(x_k)T\end{aligned}\quad (4)$$

where T denotes the sampling time. The resulting state equation can be expressed as shown in Equation (5) [26]:

$$\begin{bmatrix} i_\alpha(k+1) \\ i_\beta(k+1) \\ \omega_e(k+1) \\ \theta_e(k+1) \end{bmatrix} = \begin{bmatrix} 1 - \frac{R_s T}{L_s} & 0 & \frac{\lambda \sin \theta_e(k) T}{L_s} & \frac{\lambda \cos \theta_e(k) p \omega_r(k) T}{L_s} \\ 0 & 1 - \frac{R_s T}{L_s} & -\frac{\lambda \cos \theta_e(k) T}{L_s} & -\frac{\lambda \sin \theta_e(k) p \omega_r(k) T}{L_s} \\ -\frac{\lambda \sin \theta_e(k) T}{J} & \frac{\lambda \cos \theta_e(k) T}{J} & 1 & -(\frac{\lambda \cos \theta_e(k)}{J} i_\alpha(k) + \frac{\lambda \sin \theta_e(k)}{J} i_\beta(k)) T \\ 0 & 0 & T & 1 \end{bmatrix} \begin{bmatrix} i_\alpha(k) \\ i_\beta(k) \\ \omega_e(k) \\ \theta_e(k) \end{bmatrix} + \begin{bmatrix} \frac{T}{L_s} & 0 & 0 \\ 0 & \frac{T}{L_s} & 0 \\ 0 & 0 & \frac{T}{J} \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} U_\alpha \\ U_\beta \\ T_L \end{bmatrix} \quad (5)$$

The time-update (prediction) step estimates the a priori state and covariance matrix using the previous step's information, as shown in Equation (6):

$$\begin{aligned} \hat{x}_{pk} &= f(\hat{x}_{k-1}, u) \\ \hat{P}_{pk} &= A_k \hat{P}_{k-1} A_k^T + Q \end{aligned} \quad (6)$$

The measurement-update (correction) step refines the estimation using the current measurement data, as given in Equation (7) [30–32]:

$$\begin{aligned} K_k &= \hat{P}_{pk} H_k^T (H_k \hat{P}_{pk} H_k^T + R)^{-1} \\ \hat{x}_{ck} &= \hat{x}_{pk} + K_k (y_k - H_k (\hat{x}_{pk})) \\ \hat{P}_{ck} &= (I - K_k H_k) \hat{P}_{pk} \end{aligned} \quad (7)$$

The design and tuning of the covariance matrices Q and R play a crucial role in EKF's overall performance. These matrices determine both the dynamic response and steady-state accuracy of the estimator. In most implementations, their values are selected empirically to achieve stable operation and rapid convergence.

Assuming the functions f and h are differentiable with respect to x and u , the corresponding Jacobian matrices A_k and H_k , updated at each sampling interval, are given by Equation (8):

$$\begin{aligned} A_k &= \left. \frac{\partial f(x_k, u_k)}{\partial x} \right|_{x_k = \hat{x}_k} \\ H_k &= \left. \frac{\partial h(x_k)}{\partial x} \right|_{x_k = \hat{x}_k} \end{aligned} \quad (8)$$

4. Runge–Kutta Model-Based EKF (RKEKF)

The block diagram illustrating the sensorless operation of a PMSM using the Runge–Kutta Extended Kalman Filter (RKEKF) algorithm is shown in Figure 2. In this configuration, the three-phase current and voltage signals measured from the PMSM are first converted into their α – β components through the Clarke transformation ($a, b, c \rightarrow \alpha, \beta$). These transformed quantities are then processed by the RKEKF algorithm, which estimates the unmeasured states and provides them as feedback to the control loop.

RK5 and above) provide improved accuracy but at the cost of increased computational complexity. Since real-time performance is crucial in PMSM control applications, RK4 was chosen as it offers sufficient accuracy while maintaining feasible computational requirements for real-time implementation. Moreover, RK4 has been widely adopted in embedded control systems due to its stability and ease of implementation. The discretization process is described by Equations (13)–(15) [27]:

$$\begin{aligned}\hat{x}_1(n+1) &= \hat{x}_1(n) + \frac{1}{6}K_1X_1 + \frac{1}{3}K_2X_1 + \frac{1}{3}K_3X_1 + \frac{1}{6}K_4X_1 \\ &\vdots \\ \hat{x}_N(n+1) &= \hat{x}_N(n) + \frac{1}{6}K_1X_N + \frac{1}{3}K_2X_N + \frac{1}{3}K_3X_N + \frac{1}{6}K_4X_N\end{aligned}\quad (13)$$

and

$$\begin{aligned}\hat{y}_1(n+1) &= g_1(\hat{x}_1(n+1), \dots, \hat{x}_N(n+1), u_1(n), \dots, u_R(n)) \\ &\vdots \\ \hat{y}_Q(n+1) &= g_Q(\hat{x}_1(n+1), \dots, \hat{x}_N(n+1), u_1(n), \dots, u_R(n))\end{aligned}\quad (14)$$

where the coefficients K_1, K_2, K_3, K_4 are computed using Equation (15) as functions of the system dynamics.

$$\begin{aligned}K_1X_1 &= T_s f_1(\hat{x}_1(n), \dots, \hat{x}_N(n), u_1(n), \dots, u_R(n)) \\ &\vdots \\ K_1X_N &= T_s f_N(\hat{x}_1(n), \dots, \hat{x}_N(n), u_1(n), \dots, u_R(n)) \\ K_2X_1 &= T_s f_1(\hat{x}_1(n) + 0.5K_1X_1, \dots, \hat{x}_N(n) + 0.5K_1X_N, u_1(n), \dots, u_R(n)) \\ &\vdots \\ K_2X_N &= T_s f_N(\hat{x}_1(n) + 0.5K_1X_1, \dots, \hat{x}_N(n) + 0.5K_1X_N, u_1(n), \dots, u_R(n)) \\ K_3X_1 &= T_s f_1(\hat{x}_1(n) + 0.5K_2X_1, \dots, \hat{x}_N(n) + 0.5K_2X_N, u_1(n), \dots, u_R(n)) \\ &\vdots \\ K_3X_N &= T_s f_N(\hat{x}_1(n) + 0.5K_2X_1, \dots, \hat{x}_N(n) + 0.5K_2X_N, u_1(n), \dots, u_R(n)) \\ K_4X_1 &= T_s f_1(\hat{x}_1(n) + K_3X_1, \dots, \hat{x}_N(n) + K_3X_N, u_1(n), \dots, u_R(n)) \\ &\vdots \\ K_4X_N &= T_s f_N(\hat{x}_1(n) + K_3X_1, \dots, \hat{x}_N(n) + K_3X_N, u_1(n), \dots, u_R(n))\end{aligned}\quad (15)$$

Equations (13) and (14) can also be expressed in a more compact form, as shown in Equation (16):

$$\begin{aligned}\hat{x}(n+1) &= \hat{f}(\hat{x}(n), u(n)) \\ \hat{y}(n+1) &= g(\hat{x}(n), u(n))\end{aligned}\quad (16)$$

In this method, the matrices A_n and H_n from Equation (5) are computed using the standard RK method [27].

The updated formulations for the matrices A_n and H_n are expressed as shown in Equations (17)–(19).

$$A_n = \left. \frac{\partial \hat{f}}{\partial x} \right|_{\substack{x = \hat{x}(n-1) \\ u = u(n-1)}} \text{ and } H_n = \left. \frac{\partial g}{\partial x} \right|_{\substack{x = \hat{x}(n-1) \\ u = u(n-1)}} \quad (17)$$

here,

$$\begin{aligned}\left. \frac{\partial \hat{f}}{\partial x} \right|_{\substack{x = \hat{x}(n-1) \\ u = u(n-1)}} &= \left[\frac{\partial \hat{f}_i(\hat{x}(n-1), u(n-1))}{\partial \hat{x}_j(n-1)} \right] = \left[\frac{\partial \hat{x}_i(n)}{\partial \hat{x}_j(n-1)} \right] \\ &= \frac{\partial \hat{x}_i(n+1)}{\partial \hat{x}_j(n)} = \frac{\partial \hat{x}_i(n)}{\partial \hat{x}_j(n)} + \frac{1}{6} \frac{\partial K_1X_i(n)}{\partial \hat{x}_j(n)} + \frac{1}{3} \frac{\partial K_2X_i(n)}{\partial \hat{x}_j(n)} + \frac{1}{3} \frac{\partial K_3X_i(n)}{\partial \hat{x}_j(n)} + \frac{1}{6} \frac{\partial K_4X_i(n)}{\partial \hat{x}_j(n)}\end{aligned}\quad (18)$$

where

$$\begin{aligned}
 \frac{\partial K_1 X_i(n)}{\partial \hat{x}_j(n)} &= T_s \left[\frac{\partial f_i}{\partial x_j} \right]_{\substack{x = \hat{x}(n) \\ u = u(n)}} \\
 \frac{\partial K_2 X_i(n)}{\partial \hat{x}_j(n)} &= T_s \left[\sum_{p=1}^N \frac{\partial f_i}{\partial x_p} \left(\frac{1}{2} \frac{\partial K_1 X_p(n)}{\partial \hat{x}_j(n)} \right) \right]_{\substack{x = \hat{x}(n) + 0.5 K_1 X_i \\ u = u(n)}} \\
 \frac{\partial K_3 X_i(n)}{\partial \hat{x}_j(n)} &= T_s \left[\sum_{p=1}^N \frac{\partial f_i}{\partial x_p} \left(\frac{1}{2} \frac{\partial K_2 X_p(n)}{\partial \hat{x}_j(n)} \right) \right]_{\substack{x = \hat{x}(n) + 0.5 K_2 X_i \\ u = u(n)}} \\
 \frac{\partial K_4 X_i(n)}{\partial \hat{x}_j(n)} &= T_s \left[\sum_{p=1}^N \frac{\partial f_i}{\partial x_j} \left(\frac{\partial K_3 X_p(n)}{\partial \hat{x}_j(n)} \right) \right]_{\substack{x = \hat{x}(n) + K_3 X_i \\ u = u(n)}}
 \end{aligned} \quad (19)$$

Thus, the current state of the system can be effectively estimated using the RKEKF method [27].

5. Experimental Studies

The dSPACE DS1104 board is a standard R&D controller board designed for installation in the PCI slot of a PC. Specifically tailored for the development of high-speed, multivariable digital controllers and real-time simulations, this board enabled the real-time sensorless control of a PMSM operating at high speeds.

The DS1104 board features a TMS320F240 DSP microcontroller operating at 250 MHz. The experimental setup is illustrated in Figure 3. In the experiment, two PMSMs were used: one operated in sensorless mode, while the other acted as a coupled generator to provide a load. An inverter was also employed to implement the space vector pulse-width modulation (SVPWM) algorithm, enabling the application of the appropriate voltage to the motor windings.

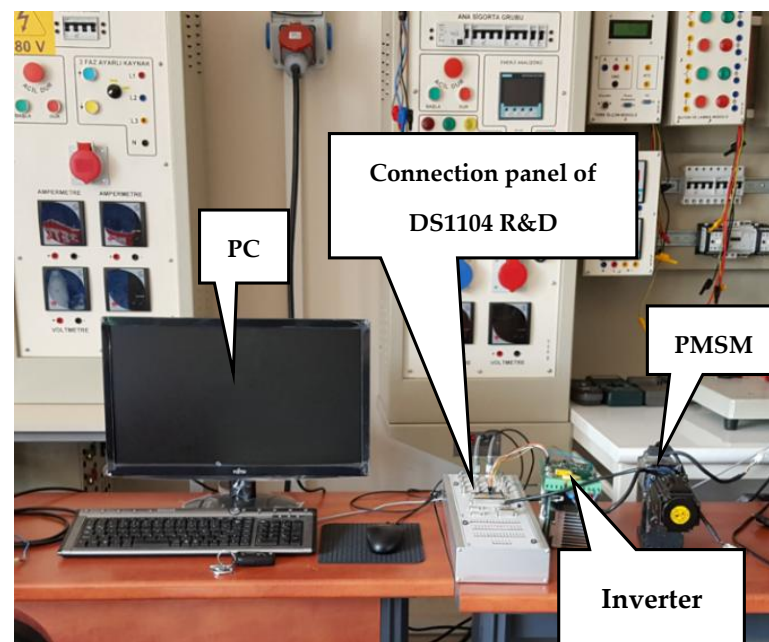


Figure 3. Experimental setup used for sensorless PMSM control with RKEKF.

The specifications of the motor used for sensorless operation are summarized in Table 1. During the experiment, the motor operated in sensorless mode, wherein the measured three-phase current and voltage signals were first transformed using the Clarke transformation. The unmeasured signals were estimated via the RKEKF algorithm and subsequently fed back as feedback signals. The rotor speed and position were derived as estimated quantities, which were then employed for the motor's speed control.

Table 1. Motor Specifications.

Parameter	Value
Rated Torque (Nm)	1.27
Rated Speed (r/min)	3000
Number of pole-pairs	4
Rated Current (A)	2.8
Stator resistance (Ω)	2.5
Stator inductance (mH)	7

Thus, with the RKEKF method, the rotor speed and position information, which is typically obtained through an encoder, was successfully estimated, enabling sensorless control of the PMSM. To ensure accurate control and synchronization with the estimation process, the PWM signals were sampled at 5 kHz.

Thus, by employing the RKEKF method, the rotor speed and position—typically measured by an encoder—were successfully estimated, enabling sensorless PMSM control.

The initial state covariance matrix P_0 , along with the model and measurement covariance matrices (Q and R , respectively), are defined as $P_0 = \text{diag}(0.1 \ 0.1 \ 0.1 \ 0.1)$, $Q = \text{diag}(0.1 \ 0.1 \ 0.1 \ 0.1)$ and $R = \text{diag}(0.001 \ 0.001)$. The process noise covariance matrix Q and the measurement noise covariance matrix R were selected as diagonal matrices to simplify tuning and reduce computational complexity. The initial values of Q were determined according to modeling uncertainties, unmodeled dynamics, and parameter variations in the PMSM, while the R matrix was chosen based on the resolution and noise characteristics of the current and voltage measurements. A trial-and-error tuning procedure was employed, starting from small covariance values and gradually increasing them until stable convergence and satisfactory estimation behavior were achieved. To ensure a fair comparison, the same Q and R matrices were used for both the EKF and the proposed RKEKF.

The PMSM was experimentally tested under several operating conditions, including forward and reverse operation, acceleration, low-speed operation, and loaded conditions. The RKEKF-based sensorless control performance was also compared with that of the conventional EKF under loaded operation.

Initially, the RKEKF was tested during sensorless forward–reverse operation. The motor was driven in the forward direction at 100 rad/s, followed by a reversal at -100 rad/s for approximately 1.8 s. At around 6.8 s, the motor was again operated forward at 100 rad/s for 10 s, and finally reversed at -100 rad/s for 4 s. As shown in Figure 4a, no overshoot occurred during the transient periods, and the actual speed closely followed the reference speed. The speed error, depicted in Figure 4b, remained nearly zero except during the directional transitions, confirming the high accuracy and robustness of the proposed RKEKF method.

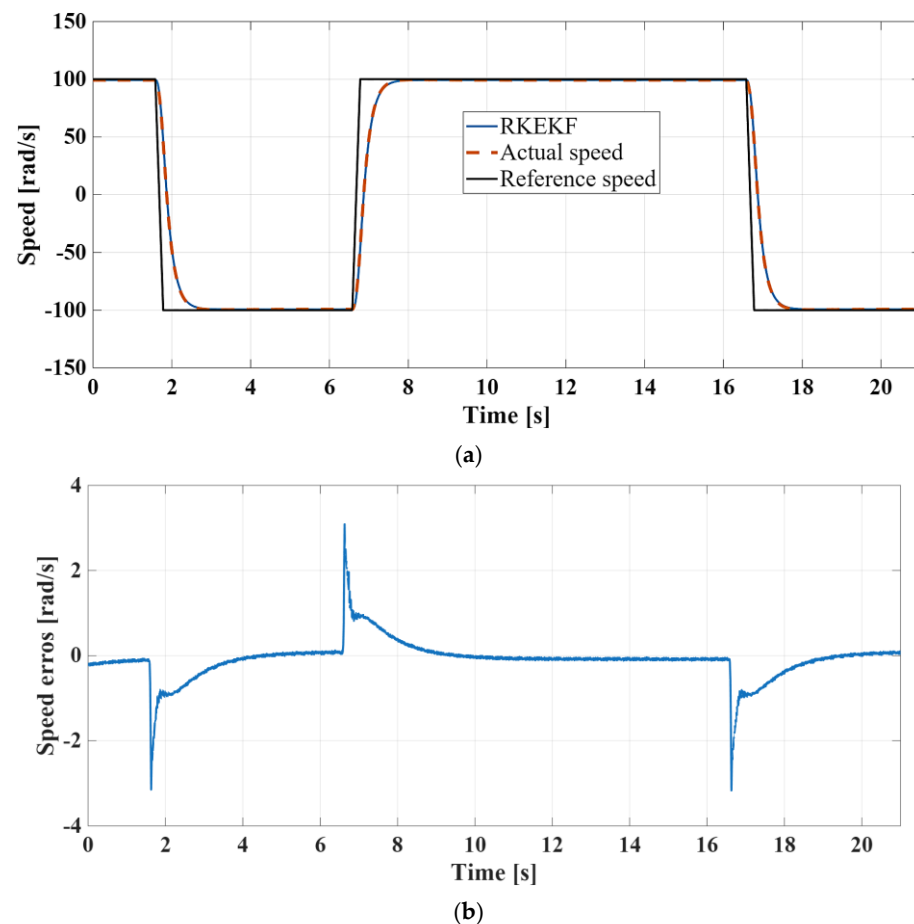


Figure 4. Experimental results of sensorless forward–reverse operation using RKEKF: (a) Speed response, (b) Speed error.

The sensorless acceleration test results are presented in Figure 5. Initially, the motor was operated at 80 rad/s for approximately 5 s, after which the reference speed was increased to 100 rad/s. As observed in Figure 5a, the actual speed closely followed the reference trajectory without overshoot during transient operation. The rotor position estimation is illustrated in Figure 5b.

Figure 6 presents the experimental low-speed response of sensorless operation with RKEKF at 60 rad/s, while Figure 7 shows the experimental high-speed response at 120 rad/s.

In Figure 6a, it is observed that both the RKEKF and the actual speed reach the reference speed of 60 rad/s quickly and smoothly in approximately 3 s. Similarly, in Figure 7a, the RKEKF and the actual speed are observed to reach the reference speed of 120 rad/s in approximately 6 s. Figures 6b and 7b show the variation in the actual and estimated positions over time ($0 \text{ s} < t < 2 \text{ s}$) as part of the transient analysis. Initially, the position estimated by the RKEKF starts with a certain error but rapidly converges to the actual position.

Figure 8 presents the experimental speed response of sensorless operation using RKEKF at 90 rad/s under full load. In Figure 8a, it is observed that both the RKEKF and the actual speed reach the reference speed quickly and smoothly in approximately 4.5 s. In Figure 8b, during the steady-state operation within the time interval of $6 \text{ s} < t < 6.5 \text{ s}$, the estimated position is seen to closely follow the actual position.

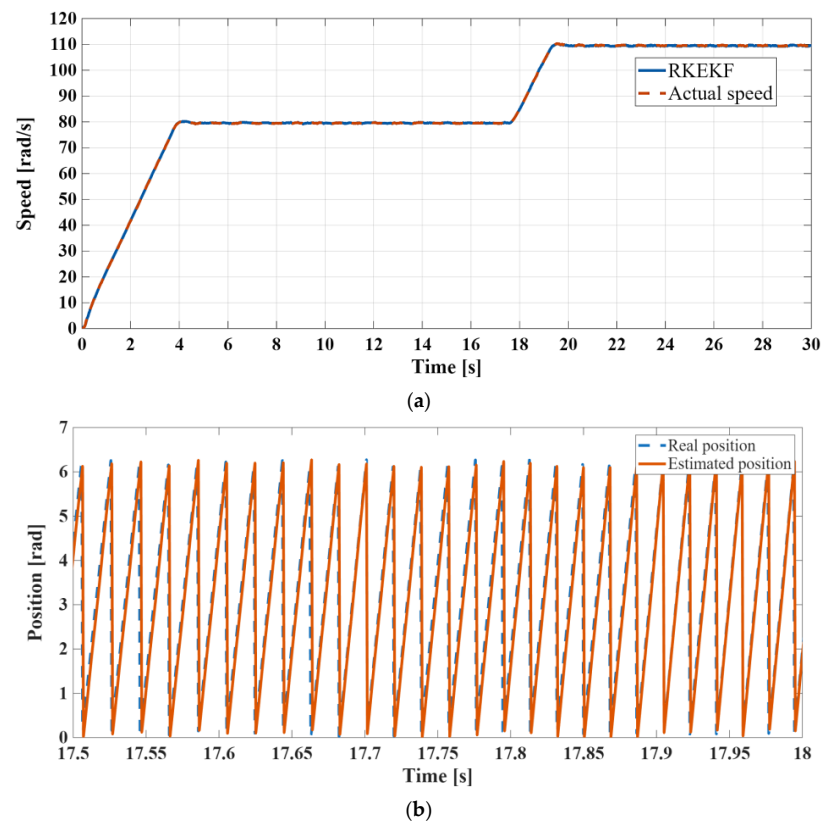


Figure 5. Sensorless acceleration performance of PMSM with RKEKF: (a) Speed response, (b) Rotor position estimation.

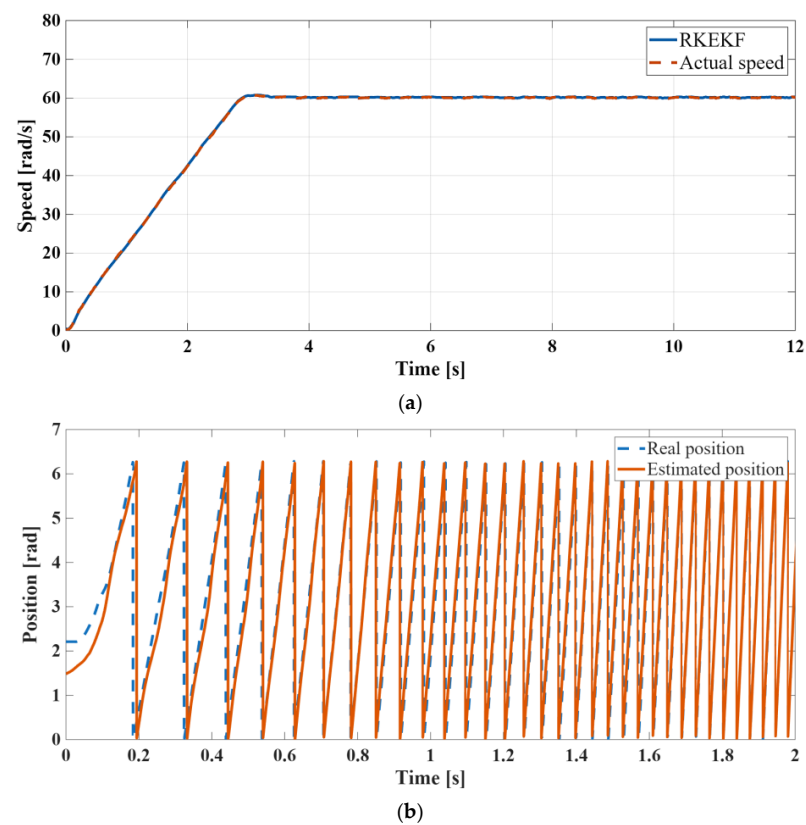


Figure 6. Experimental low-speed response of sensorless operation using RKEKF at 60 rad/s. (a) Speed response (b) Rotor position estimation.

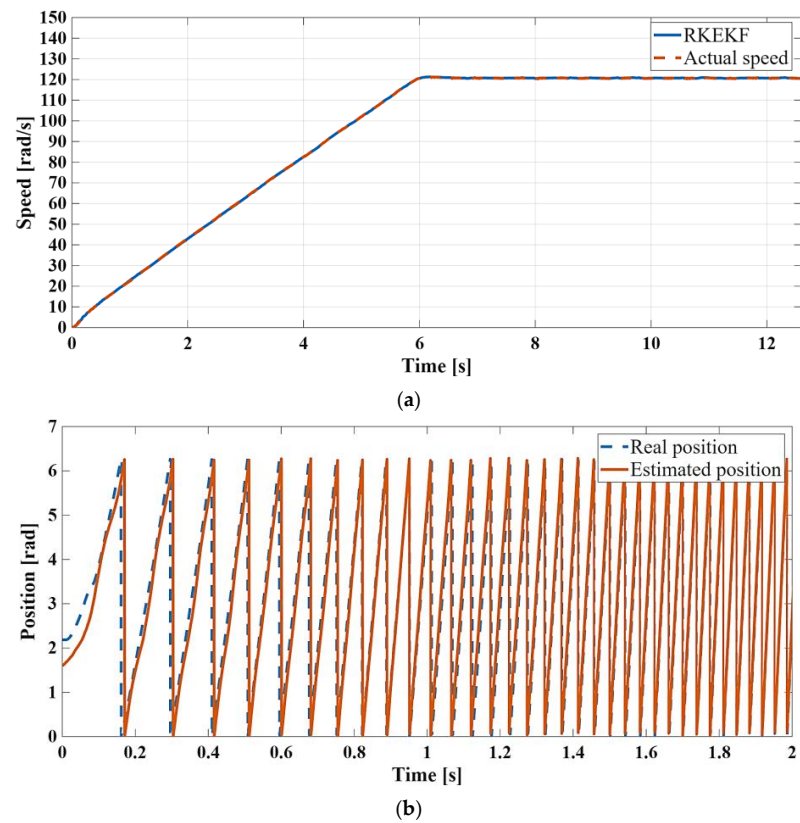


Figure 7. Experimental high-speed response of sensorless operation using RKEKF at 120 rad/s. (a) Speed response (b) Rotor position estimation.

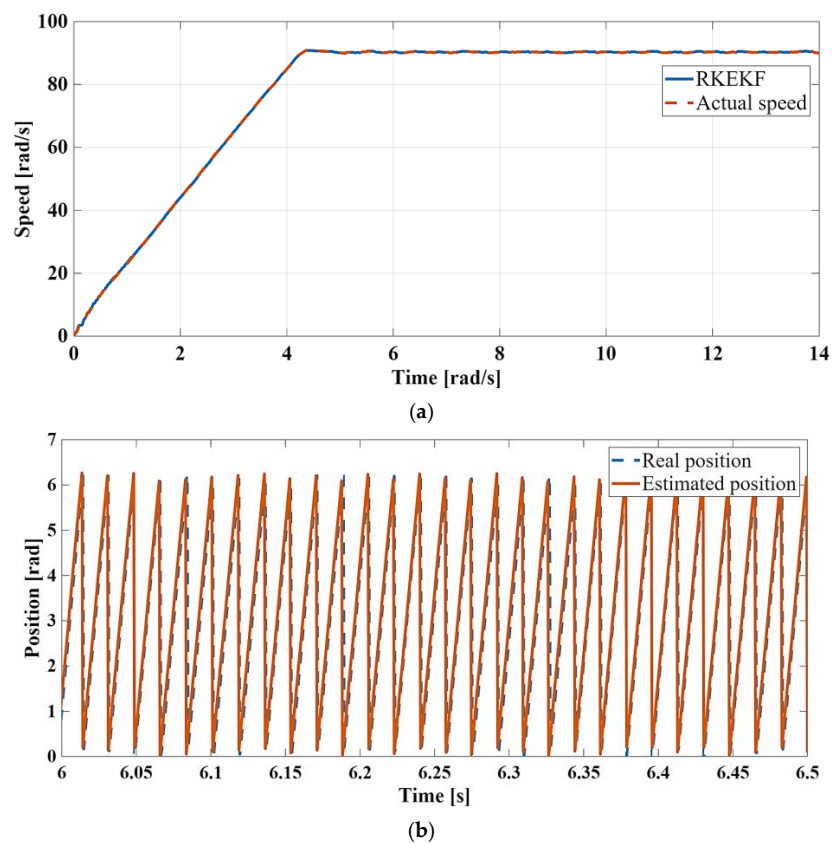


Figure 8. Experimental speed response of the sensorless operation with RKEKF under loaded conditions: (a) Speed response; (b) Rotor position estimation.

Finally, the performances of both the RKEKF and EKF under loaded operation were evaluated and compared, as illustrated in Figure 9. The tests were conducted at a speed of 80 rad/s under load. In both cases, the reference speed was accurately tracked without overshoot during the transient state, demonstrating the effectiveness of the proposed RKEKF method. In this figure, the RKEKF demonstrates a quicker convergence to the steady-state value compared to the EKF, which indicates improved computational efficiency and adaptability in dynamic conditions.

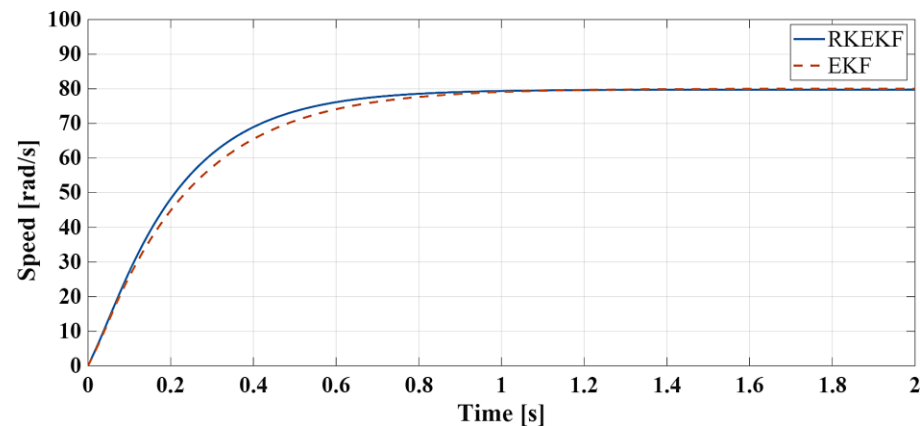


Figure 9. Comparison of experimental speed responses between RKEKF and EKF in sensorless operation under loaded conditions.

Additionally, to evaluate the real-time applicability of the proposed RKEKF algorithm, the execution times of both the conventional EKF and the RKEKF were experimentally measured on the dSPACE DS1104 platform. While the EKF algorithm exhibited an execution time of 47 μ s, the RKEKF algorithm demonstrated an execution time of 55 μ s. Although RKEKF entails a higher computational cost due to the increased number of integration steps, it offers superior estimation accuracy and smoother convergence characteristics. Despite this additional computational load, the execution time of the RKEKF remains within the limits required for real-time PMSM control applications. Although experimental validation in this study is limited to low- and medium-speed operating regions, the measured execution time confirms that the proposed RKEKF can be executed within a single control cycle without violating real-time constraints. Since the estimator performance is primarily constrained by computational execution time rather than mechanical speed itself, these results indicate that the proposed RKEKF-based sensorless control framework is computationally suitable for extension to high-speed and rated-speed PMSM operation, which will be addressed in future experimental studies. Furthermore, the algorithm's structure and computational profile also make it suitable for implementation on other high-speed platforms, such as Digital Signal Processors (DSPs).

6. Conclusions

This study focused on the sensorless control of PMSMs, which are widely utilized across many fields due to their favorable dynamic and efficiency characteristics. Although these machines are commonly operated with feedback devices such as encoders or resolvers to ensure accurate speed and position information, such sensors inevitably raise the overall cost and impose additional size and integration constraints. These limitations have driven the increasing interest in sensorless estimation techniques.

In this work, a sensorless PMSM control framework based on an Extended Kalman Filter formulation incorporating the Runge–Kutta model was developed and experimentally validated. The proposed approach was tested on the dSPACE DS1104 platform under

four distinct operating scenarios: forward and reverse rotation, acceleration, low-speed operation, and operation under load. For the loaded condition, the classical EKF was also implemented to enable performance comparison with the RKEKF method.

The results reveal that the RKEKF provides rapid and overshoot-free convergence to the reference speed during transients. The adoption of the RK4 numerical integration procedure offers a favorable compromise between computational demand and estimation accuracy, allowing real-time implementation without degrading the control response.

While the fast transient convergence of the RKEKF reflects its estimation quality, computational efficiency is more directly indicated by execution time. Experimental measurements showed an execution time of 55 μ s, demonstrating that the proposed algorithm satisfies real-time requirements. Furthermore, the computational structure of the estimator is well-suited to high-speed hardware such as DSP-based systems.

Overall, the findings of this research identify the RKEKF approach as a robust and effective solution for sensorless PMSM control, particularly in industrial environments. In this study, the estimator design intentionally focuses on direct estimation of rotor speed and position without explicitly including back-electromotive force as an additional state variable. Although direct back-EMF estimation can enhance robustness against permanent-magnet flux linkage variations, it also increases model complexity and computational burden, which may adversely affect real-time performance. Since the primary objective of this work is to improve numerical integration accuracy and real-time feasibility of EKF-based sensorless control, the state vector was deliberately kept compact. In future studies, quantitative error-based performance metrics, such as RMSE, will be incorporated to further assess estimation accuracy. In addition, the robustness of the proposed RKEKF under varying operating conditions and parameter uncertainties will be investigated. Another important research direction involves the development of adaptive or self-tuning strategies for the noise covariance matrices to further enhance estimation performance while preserving real-time feasibility. Future studies may also investigate extended RKEKF formulations that explicitly incorporate back-EMF estimation and parameter variation compensation, particularly in applications where advanced hardware resources mitigate computational limitations.

Funding: This research received no external funding.

Data Availability Statement: The original contributions presented in this study are included in the article. Further inquiries can be directed to the corresponding author.

Conflicts of Interest: The author declares no conflict of interest.

References

1. Yi, B.; Kang, L.; Tao, S.; Zhao, X.; Jing, Z. Adaptive Two-Stage Extended Kalman Filter Theory in Application of Sensorless Control for Permanent Magnet Synchronous Motor. *Math. Probl. Eng.* **2013**, *2013*, 974974. [[CrossRef](#)]
2. Akpunar, A.; İplikçi, S. Runge-Kutta Model Predictive Speed Control for Permanent Magnet Synchronous Motors. *Energies* **2020**, *13*, 1216. [[CrossRef](#)]
3. Akpunar, A.; İplikçi, S. Control and Parameter Estimation of PMSM by Runge-Kutta Model Based Predictive Control. In Proceedings of the 12th International Symposium on Advanced Topics in Electrical Engineering (ATEE 2021), Bucharest, Romania, 25–27 March 2021. [[CrossRef](#)]
4. Wang, A.; Wei, S. Sliding Mode Control for Permanent Magnet Synchronous Motor Drive Based on an Improved Exponential Reaching Law. *IEEE Access* **2019**, *7*, 146866–146875. [[CrossRef](#)]
5. Li, S.; Wang, H.; Li, H.; Yang, C. PMSM Sliding Mode Control Based on Novel Reaching Law and Extended State Observer. *Adv. Mech. Eng.* **2022**, *14*, 1–16. [[CrossRef](#)]
6. Li, S.; Gu, H. Fuzzy Adaptive Internal Model Control Schemes for PMSM Speed-Regulation System. *IEEE Trans. Ind. Inform.* **2012**, *8*, 767–779. [[CrossRef](#)]

7. Li, S.; Liu, Z. Adaptive Speed Control for Permanent-Magnet Synchronous Motor System With Variations of Load Inertia. *IEEE Trans. Ind. Electron.* **2009**, *56*, 3050–3059. [\[CrossRef\]](#)
8. Asseu, O.; Yoboué, P.; Konaté, A.; Diaby, M. An Extended Kalman Filter Algorithm for A PMSM: Experimental Results. *Asian J. Appl. Sci.* **2017**, *10*, 32–38. [\[CrossRef\]](#)
9. Kung, Y.-S.; Vu Quynh, N.; Trung Hieu, N.; Lin, J.-M. FPGA Realization of Sensorless PMSM Speed Controller Based on Extended Kalman Filter. *Math. Probl. Eng.* **2013**, *2013*, 919318. [\[CrossRef\]](#)
10. Wang, Q.; Wang, S.; Chen, C. Review of Sensorless Control Techniques for PMSM Drives. *IEEJ Trans. Electr. Electron. Eng.* **2019**, *14*, 1543–1552. [\[CrossRef\]](#)
11. Khadija, K.; Benyounès, M.; Khalil, B.I.; Rachid, B.M. A simple and robust Speed Tracking Control of PMSM. *Przegląd Elektrotechniczny* **2011**, *87*, 202–207.
12. Li, H.; Ke, D.; Zu, R.; Tao, P.; Wang, F. Sensorless Control of Permanent Magnet Synchronous Motor Based on An Improved Sliding Mode Observer. In Proceedings of the 2018 IEEE Student Conference on Electric Machines and Systems, Huzhou, China, 14–16 December 2018. [\[CrossRef\]](#)
13. Adamu, S.S.; Lawan, M.B. Permanent Magnet Synchronous Motor Rotor Position Estimation Using Fuzzy-Based Sliding Mode Observer. In Proceedings of the 2017 9th IEEE-GCC Conference and Exhibition (GCCCE), Manama, Bahrain, 8–11 May 2017. [\[CrossRef\]](#)
14. Feifei, H.; Zhonghua, W.; Yueyang, L.; Tongyi, H. Sensorless speed control of permanent magnet synchronous motor based on RBF neural network. In Proceedings of the 2015 34th Chinese Control Conference (CCC), Hangzhou, China, 28–30 July 2015. [\[CrossRef\]](#)
15. Abo-Khalil, A.G.; Eltamaly, A.M.; Alsaud, M.S.; Sayed, K.; Alghamdi, A.S. Sensorless control for PMSM using model reference adaptive system. *Int. Trans. Electr. Energy Syst.* **2020**, *31*, e12733. [\[CrossRef\]](#)
16. Allaoui, S.; Laamari, Y.; Chafaa, L.; Saad, S. Position and Speed Estimation of PMSM Based On Extended Kalman Filter Tuned by Biogeography-Based-Optimization. *J. Eur. Syst. Autom.* **2021**, *54*, 559–568. [\[CrossRef\]](#)
17. Tety, P.; Konaté, A.; Asseu, O.; Soro, E.; Yoboué, P.; Kouadjo, A.R. A Robust Extended Kalman Filter for Speed-Sensorless Control of a Linearized and Decoupled PMSM Drive. *Engineering* **2015**, *7*, 691–699. [\[CrossRef\]](#)
18. Elahi, U.; Hassan, M.; Kashif, S.A.R. Sensorless Control of A Permanent Magnet Synchronous Motor Using A Novel Four Switch Three Phase Inverter and Extended Kalman Filter. *Sci. Int.* **2015**, *27*, 1861–1868.
19. Chen, Z.; Liu, Y. Sensorless control of marine permanent magnet synchronous propulsion motor based on adaptive extended Kalman filter. *Front. Energy Res.* **2022**, *10*, 1037595. [\[CrossRef\]](#)
20. Dilys, J.; Stankevič, V.; Łuksza, K. Implementation of Extended Kalman Filter with Optimized Execution Time for Sensorless Control of a PMSM Using ARM Cortex-M3 Microcontroller. *Energies* **2021**, *14*, 3491. [\[CrossRef\]](#)
21. Jang, J.-S.; Park, B.-G.; Kim, T.-S.; Lee, D.M.; Hyun, D.-S. Sensorless Control of Four-Switch Three-Phase PMSM Drive Using Extended Kalman Filter. In Proceedings of the 2008 34th Annual Conference of IEEE Industrial Electronics, Orlando, FL, USA, 10–13 November 2008. [\[CrossRef\]](#)
22. Megrini, M.; Gaga, A.; Mehdaoui, Y.; Khyat, J. Design and PIL test of extended Kalman filter for PMSM field oriented control. *Results Eng.* **2024**, *24*, 102843. [\[CrossRef\]](#)
23. Than, H.; Kung, Y.S.; Risfendra, R. Realization of a speed controller for sensorless PMSM drives using LFSI and EKF. In *Engineering Innovation and Design*; CRC Press: Boca Raton, FL, USA, 2019; pp. 98–104.
24. Quang, N.K.; Tung, D.D.; Ha, Q.P. FPGA-based sensorless PMSM speed control using adaptive extended Kalman filter. In Proceedings of the 2015 IEEE International Conference on Automation Science and Engineering (CASE), Gothenburg, Sweden, 24–28 August 2015. [\[CrossRef\]](#)
25. Terorde, G. Sensorless control of a permanent magnet synchronous motor for PV-powered water pump systems using the extended Kalman filter. In Proceedings of the 9th International Conference on Electrical Machines and Drives, Canterbury, UK, 1–3 September 1999. [\[CrossRef\]](#)
26. Sun, P.; Chen, B.; Zheng, S.; Deng, L.; Wei, G.; Wang, H. Sensorless Control of SPMSM with Improved Extended Kalman Filter Based on Runge–Kutta Model. In Proceedings of the 27th International Conference on Electrical Machines and Systems (ICEMS 2024), Fukuoka, Japan, 26–29 November 2024; pp. 188–192. [\[CrossRef\]](#)
27. Iplikci, S. Runge-Kutta model-based adaptive predictive control mechanism for nonlinear processes. *Trans. Inst. Meas. Control* **2013**, *35*, 166–180. [\[CrossRef\]](#)
28. Tang, M.; Gao, L.; Liang, D.; Xian, J.T. Sensorless Permanent Magnet Synchronous Motor drive using an optimized and normalized Extended Kalman filter. In Proceedings of the 2011 International Conference on Electrical Machines and Systems, Beijing, China, 20–23 August 2011; pp. 1–4. [\[CrossRef\]](#)
29. Borsje, P.; Chan, T.F.; Wong, Y.K.; Ho, S.L. A Comparative Study of Kalman Filtering for Sensorless Control of a Permanent-Magnet Synchronous Motor Drive. In Proceedings of the IEEE International Conference on Electric Machines and Drives 2005, San Antonio, TX, USA, 15 May 2005. [\[CrossRef\]](#)

30. Gowda, M.; Ali, W.H.; Cofie, P.; Fuller, J. Design and Digital Implementation of Controller for PMSM Using Extended Kalman Filter. *Circuits Syst.* **2013**, *4*, 489–497. [[CrossRef](#)]
31. Peicheng, S.; Chen, W.; Rongyun, Z.; Suo, W. Research on Sensorless Control Method of Permanent Magnet Synchronous Motor Based on Iterated Cubature Kalman Filter. *Sci. Prog.* **2019**, *103*, 1–24. [[CrossRef](#)] [[PubMed](#)]
32. Urbanski, K.; Janiszewski, D. Sensorless Control of the Permanent Magnet Synchronous Motor. *Sensors* **2019**, *19*, 3546. [[CrossRef](#)] [[PubMed](#)]

Disclaimer/Publisher’s Note: The statements, opinions and data contained in all publications are solely those of the individual author(s) and contributor(s) and not of MDPI and/or the editor(s). MDPI and/or the editor(s) disclaim responsibility for any injury to people or property resulting from any ideas, methods, instructions or products referred to in the content.